SEEING CHANGE FROM ORBIT · MEETING MINUTES

Meeting 5: SAM Segmentation Results

Date and time	Mon 5 Oct 2026, 1:30–2:30 pm
Type	Supervisor
Location	Online (MS Teams)
Attendees	Dr. Panos, Khaled, Zain, Ayesha, Tehsin, Zayed

Agenda

Topics covered:

1. Ayesha's SAM mangrove segmentation results
2. HPC access
3. Model choices, compute and ground truth
4. Gantt chart and follow-up

Member updates

Who	Update
Ayesha	SAM ViT-H mangrove segmentation report (3 Oct), presented in the meeting
Tehsin	Eastern Mangroves 2023/2025 NDVI/NDWI exploratory comparison (30 Sep); 6-band Sentinel-2 exports for 10 Dec 2018 and 2025, documented with checksums (3 Oct); Track A alignment and 2-vs-3 landmark registration-error experiment on the Jubail pair, validated on the real 6-band rasters (4 Oct)
Khaled	Track A landmark detection and detector comparison (4 Oct)

Discussion

1. SAM segmentation results (Ayesha)

- Report shared beforehand: SAM ViT-H, frozen encoder, decoder fine-tuned, GMW 2020 labels, NDVI/NDWI as mask prompts.
- Test IoU rose from 0.09 (no prompt) to 0.36 with the spectral indices; a plain NDVI threshold scored slightly higher (IoU 0.41).
- Dr. Panos: the indices drive most of the gain; questioned how much SAM's features contribute, the frozen encoder's runtime, and signs of overfitting. Establish a U-Net baseline.

2. HPC access

- Dr. Panos offered the university HPC while other compute is being arranged and walked Ayesha through the user request. Approval and research-computing training are required first.

3. Model choices, compute and ground truth

- Limited compute; online GPUs are an option. The team plans to compare several models.

- Dr. Panos suggested models pretrained on multispectral data, and moving from 25 m to 10 m ground truth (affects performance and interpretation).

4. Gantt chart and follow-up

- Ayesha asked Dr. Panos for written action items. Dr. Panos: he can't micromanage; what helps is an agenda, a summary of what has been done, and the updated Gantt chart. He said the written updates sent before the meeting were useful.
- Tehsin shared progress with Dr. Panos in writing (Track A alignment and the 2-vs-3 landmark experiment, Eastern Mangroves comparison, 6-band exports).
- He only had the old high-level plan (multispectral fix, data dictionary, finer data + ADIA). He will review the Read AI notes from the last two meetings with the new chart and send written feedback.

Decisions

- Add a U-Net baseline to compare against SAM.
- Consider multispectral-pretrained models and 10 m ground truth.
- The team sends a progress summary and the updated Gantt chart with each agenda.

Action items

Who	What	Due
Ayesha	Complete research-computing training; submit HPC user request	Not set
Dr. Panos	Approve Ayesha's HPC request	On receiving the email
Ayesha, team	Refine the Gantt chart and send it to Dr. Panos	Not set
Dr. Panos	Review recent meeting notes and the updated Gantt chart; send written feedback	After receiving the chart

Next meeting

Not recorded. No records exist for 12 or 19 Oct.